

INA226 高侧/低侧测量、 双向电流/功率监视器，具有 I²C 兼容接口

1 特性

- 感测的总线电压范围：0V 至 36V
- 高侧或低侧感测
- 报告电流、电压和功率
- 高精度：
 - 0.1% 增益误差（最大值）
 - 10 μ V 偏移（最大值）
- 可配置的取平均值选项
- 16 个可编程地址
- 由 2.7V 至 5.5V 电源供电
- 10 引脚 DGS 超薄小外形尺寸 (VSSOP) 封装

2 应用范围

- 服务器
- 电信设备
- 计算
- 电源管理
- 电池充电器
- 电源
- 测试设备

3 说明

INA226 是一款分流/功率监视器，具有 I²C™ 或 SMBUS 兼容接口。该器件监视分流压降和总线电源电压。可编程校准值、转换时间和取平均值功能与内部乘法器相结合，可实现电流值（单位为安培）和功率值（单位为瓦）的直接读取。

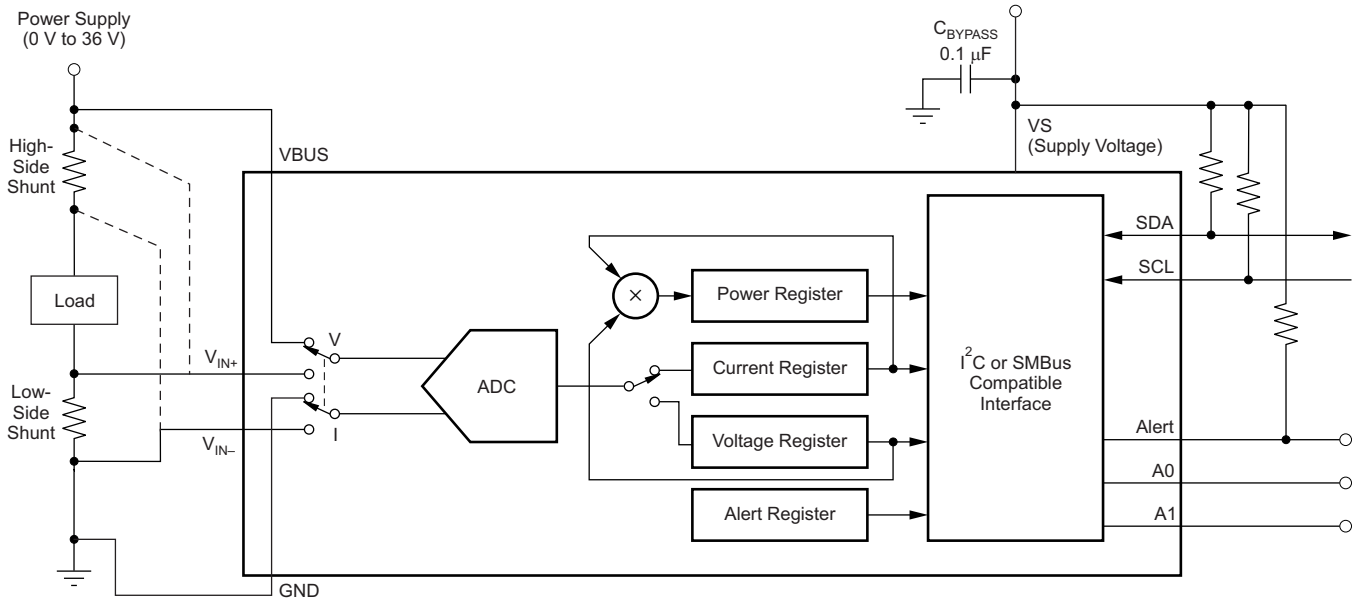
INA226 可在 0V 至 36V 的共模总线电压范围内感测电流，与电源电压无关。该器件由一个 2.7V 至 5.5V 的单电源供电，电源电流典型值为 330 μ A。该器件的额定工作温度范围为 -40°C 至 125°C，I²C 兼容接口上具有多达 16 个可编程地址。

器件信息⁽¹⁾

器件型号	封装	封装尺寸（标称值）
INA226	VSSOP (10)	3.00mm x 3.00mm

(1) 如需了解所有可用封装，请见数据表末尾的可订购产品附录。

高侧或低侧感测应用



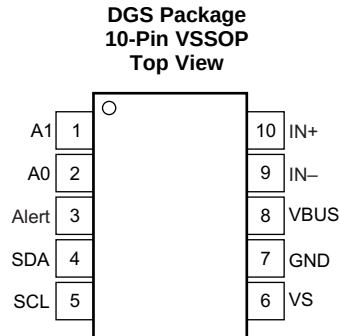
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4 修订历史记录

Changes from Original (June 2011) to Revision A	Page
已添加 处理额定值表，特性描述部分，器件功能模式，应用和实施部分，电源相关建议部分，布局部分，器件和文档支持部分以及机械、封装和可订购信息部分	1

5 Pin Configuration and Functions



Pin Functions

PIN		I/O	DESCRIPTION
NAME	NO.		
A0	2	Digital input	Address pin. Connect to GND, SCL, SDA, or VS. 表 2 shows pin settings and corresponding addresses.
A1	1	Digital input	Address pin. Connect to GND, SCL, SDA, or VS. 表 2 shows pin settings and corresponding addresses.
Alert	3	Digital output	多功能报警，开漏输出。
GND	7	Analog	Ground.
IN+	10	Analog input	连接至分流电阻的电源侧
IN-	9	Analog input	连接至分流电阻的负载侧
SCL	5	Digital input	串行总线时钟线，开漏输入
SDA	4	Digital I/O	串行总线数据线，开漏输入/输出。
VBUS	8	Analog input	总线电压输入。
VS	6	Analog	电源，2.7V 至 5.5V。

6 Specifications

6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted) ⁽¹⁾

		MIN	MAX	UNIT
V _{VS}	Supply voltage		6	V
Analog Inputs, IN+, IN–	Differential (V _{IN+} – V _{IN–}) ⁽²⁾	–40	40	V
	Common-Mode (V _{IN+} + V _{IN–}) / 2	–0.3	40	
V _{VBUS}		–0.3	40	V
V _{SDA}		GND – 0.3	6	V
V _{SCL}		GND – 0.3	V _{VS} + 0.3	V
I _{IN}	Input current into any pin		5	mA
I _{OUT}	Open-drain digital output current		10	mA
T _J	Junction temperature		150	°C
T _{stg}	Storage temperature range	–65	150	°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) IN+ and IN– may have a differential voltage between –40 V and 40 V. However, the voltage at these pins must not exceed the range –0.3 V to 40 V.

6.2 ESD Ratings

		VALUE	UNIT
V _(ESD)	Electrostatic discharge		
	Human body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins ⁽¹⁾	±2500	V
	Charged device model (CDM), per JEDEC specification JESD22-C101, all pins ⁽²⁾	±1000	
	Machine model (MM)	±150	

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

		MIN	NOM	MAX	UNIT
V _{CM}	Common-mode input voltage		12		V
V _{VS}	Operating supply voltage		3.3		V
T _A	Operating free-air temperature	–40		125	°C

6.4 Thermal Information

THERMAL METRIC ⁽¹⁾		INA226	UNIT
		DGS (VSSOP)	
		10 PINS	
R _{θJA}	Junction-to-ambient thermal resistance	171.4	°C/W
R _{θJC(top)}	Junction-to-case (top) thermal resistance	42.9	°C/W
R _{θJB}	Junction-to-board thermal resistance	91.8	°C/W
Ψ _{JT}	Junction-to-top characterization parameter	1.5	°C/W
Ψ _{JB}	Junction-to-board characterization parameter	90.2	°C/W
R _{θJC(bot)}	Junction-to-case (bottom) thermal resistance	n/a	°C/W

- (1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).

6.5 Electrical Characteristics

$T_A = 25^\circ\text{C}$, $V_{VS} = 3.3\text{ V}$, $V_{IN+} = 12\text{ V}$, $V_{SENSE} = (V_{IN+} - V_{IN-}) = 0\text{ mV}$ and $V_{VBUS} = 12\text{ V}$, unless otherwise noted

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT
INPUT						
	Shunt voltage input range		−81.9175		81.92	mV
	Bus voltage input range ⁽¹⁾		0		36	V
CMRR	Common-mode rejection	0 V ≤ V _{IN+} ≤ 36 V	126	140		dB
V _{OS}	Shunt offset voltage, RTI ⁽²⁾			±2.5	±10	μV
	Shunt offset voltage, RTI ⁽²⁾ vs temperature	−40°C ≤ T _A ≤ 125°C		0.02	0.1	μV/°C
PSRR	Shunt offset voltage, RTI ⁽²⁾ vs Power supply	2.7 V ≤ VS ≤ 5.5 V		2.5		μV/V
V _{OS}	Bus offset voltage, RTI ⁽²⁾			±1.25	±7.5	mV
	Bus offset voltage, RTI ⁽²⁾ vs temperature	−40°C ≤ T _A ≤ 125°C		10	40	μV/°C
PSRR	Bus offset voltage, RTI ⁽²⁾ vs power supply			0.5		mV/V
I _B	Input bias current (I _{IN+} , I _{IN−} pins)			10		μA
	VBUS input impedance			830		kΩ
	Input leakage ⁽³⁾	(IN+ pin) + (IN− pin), Power-down mode		0.1	0.5	μA
DC ACCURACY						
	ADC native resolution			16		Bits
	1 LSB step size	Shunt voltage		2.5		μV
		Bus voltage		1.25		mV
	Shunt voltage gain error			0.02%	0.1%	
	Shunt voltage gain error vs temperature	−40°C ≤ T _A ≤ 125°C		10	50	ppm/°C
	Bus voltage gain error			0.02%	0.1%	
	Bus voltage gain error vs temperature	−40°C ≤ T _A ≤ 125°C		10	50	ppm/°C
	Differential nonlinearity			±0.1		LSB
t _{CT}	ADC conversion time	CT bit = 000		140	154	μs
		CT bit = 001		204	224	
		CT bit = 010		332	365	
		CT bit = 011		588	646	
		CT bit = 100		1.1	1.21	ms
		CT bit = 101		2.116	2.328	
		CT bit = 110		4.156	4.572	
		CT bit = 111		8.244	9.068	
SMBus						
	SMBus timeout ⁽⁴⁾			28	35	ms

- (1) While the input range is 36 V, the full-scale range of the ADC scaling is 40.96 V. See the [Basic ADC Functions](#) section. Do not apply more than 36 V.
- (2) RTI = Referred-to-input.
- (3) Input leakage is positive (current flowing into the pin) for the conditions shown at the top of this table. Negative leakage currents can occur under different input conditions.
- (4) SMBus timeout in the INA226 resets the interface any time SCL is low for more than 28 ms.

Electrical Characteristics (接下页)
 $T_A = 25^\circ\text{C}$, $V_{VS} = 3.3\text{ V}$, $V_{IN+} = 12\text{ V}$, $V_{SENSE} = (V_{IN+} - V_{IN-}) = 0\text{ mV}$ and $V_{VBUS} = 12\text{ V}$, unless otherwise noted

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
DIGITAL INPUT/OUTPUT					
Input capacitance			3		pF
Leakage input current	$0\text{ V} \leq V_{SCL} \leq V_{VS}$, $0\text{ V} \leq V_{SDA} \leq V_{VS}$, $0\text{ V} \leq V_{Alert} \leq V_{VS}$, $0\text{ V} \leq V_{A0} \leq V_{VS}$, $0\text{ V} \leq V_{A1} \leq V_{VS}$		0.1	1	μA
V_{IH} High-level input voltage		$0.7 \times V_{VS}$		6	V
V_{IL} Low-level input voltage		-0.5		$0.3 \times V_{VS}$	V
V_{OL} Low-level output voltage, SDA, Alert	$I_{OL} = 3\text{ mA}$	0		0.4	V
Hysteresis			500		mV
POWER SUPPLY					
Operating supply range		2.7		5.5	V
I_Q Quiescent current			330	420	μA
Quiescent current, power-down (shutdown) mode			0.5	2	μA
V_{POR} Power-on reset threshold			2		V

6.6 Typical Characteristics

At $T_A = 25^\circ\text{C}$, $V_{VS} = 3.3\text{ V}$, $V_{IN+} = 12\text{ V}$, $V_{SENSE} = (V_{IN+} - V_{IN-}) = 0\text{ mV}$ and $V_{VBUS} = 12\text{ V}$, unless otherwise noted.

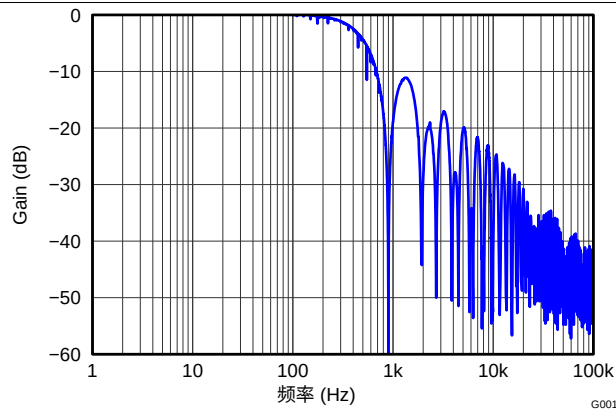


图 1. 频率响应

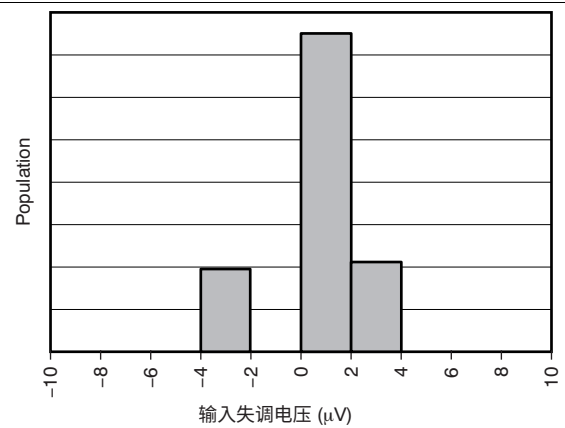


图 2. 分流输入失调电压生产分布

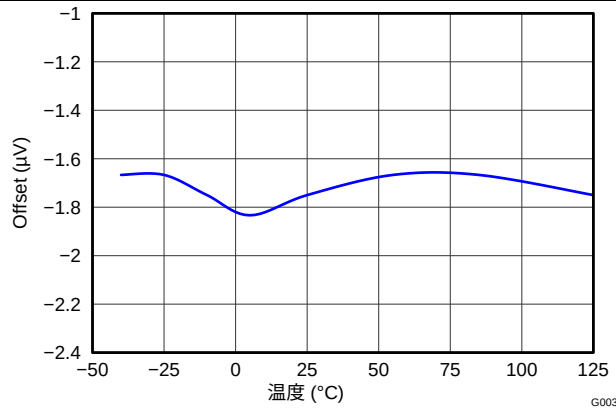


图 3. Shunt Input Offset Voltage vs Temperature

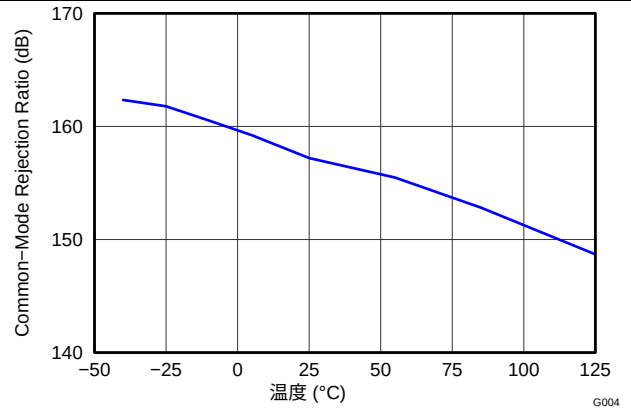


图 4. Shunt Input Common-Mode Rejection Ratio vs Temperature

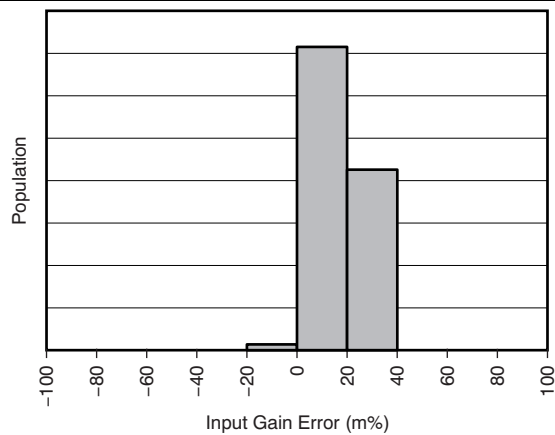


图 5. Shunt Input Gain Error Production Distribution

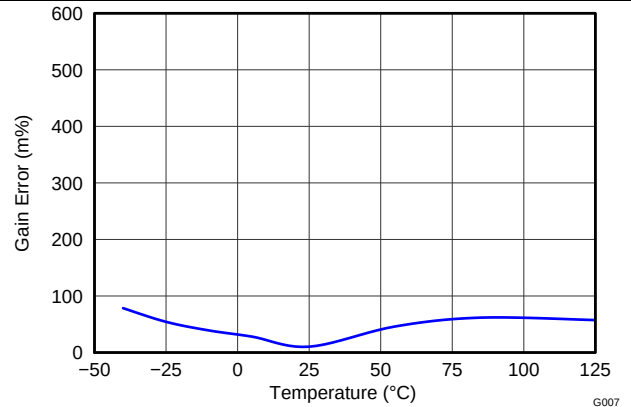


图 6. Shunt Input Gain Error vs Temperature

Typical Characteristics (接下页)

At $T_A = 25^\circ\text{C}$, $V_{VS} = 3.3\text{ V}$, $V_{IN+} = 12\text{ V}$, $V_{SENSE} = (V_{IN+} - V_{IN-}) = 0\text{ mV}$ and $V_{VBUS} = 12\text{ V}$, unless otherwise noted.

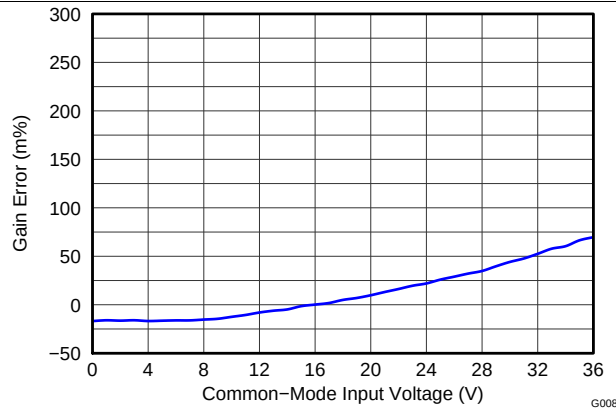


图 7. 分流输入增益误差与共模电压的关系

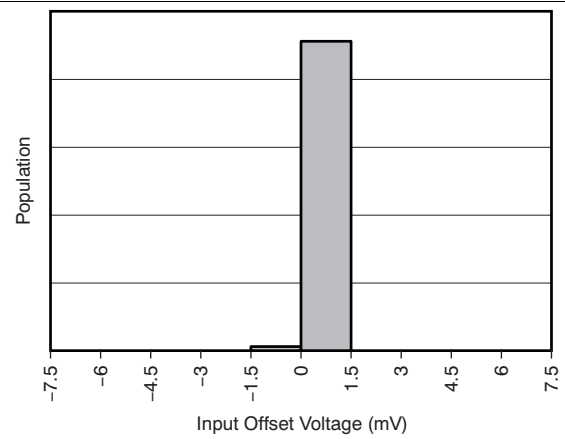


图 8. 总线输入失调电压生产分布

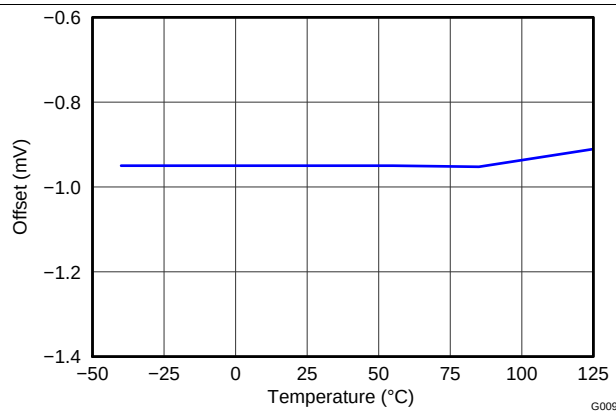


图 9. 总线输入失调电压与温度的关系

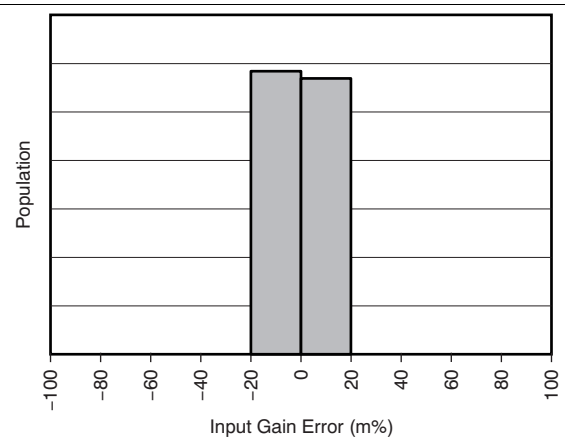


图 10. 总线输入增益误差生产分布

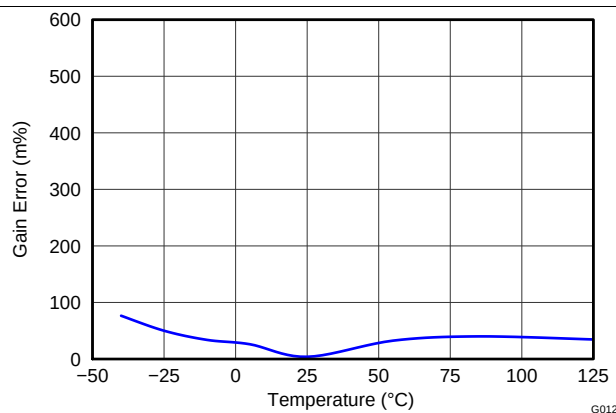


图 11. Bus Input Gain Error vs Temperature

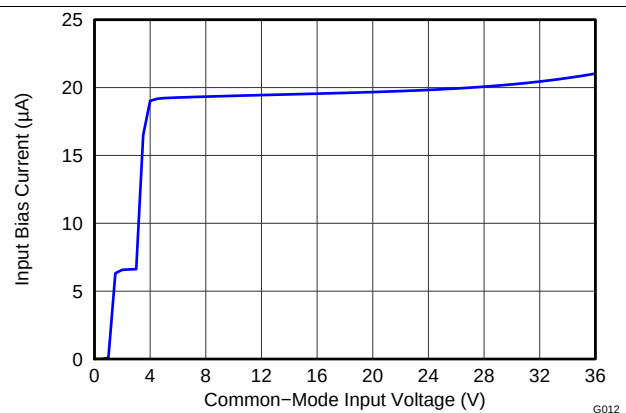


图 12. Input Bias Current vs Common-Mode Voltage

Typical Characteristics (接下页)

At $T_A = 25^\circ\text{C}$, $V_{VS} = 3.3\text{ V}$, $V_{IN+} = 12\text{ V}$, $V_{SENSE} = (V_{IN+} - V_{IN-}) = 0\text{ mV}$ and $V_{VBUS} = 12\text{ V}$, unless otherwise noted.

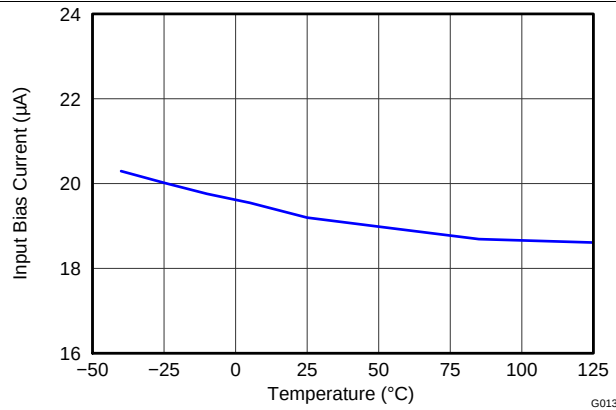


图 13. Input Bias Current vs Temperature

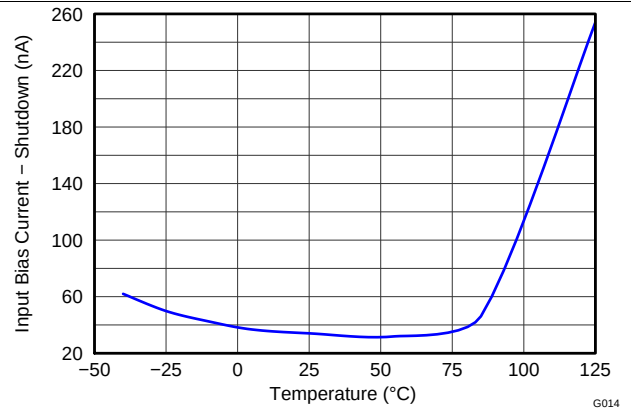


图 14. Input Bias Current vs Temperature, Shutdown

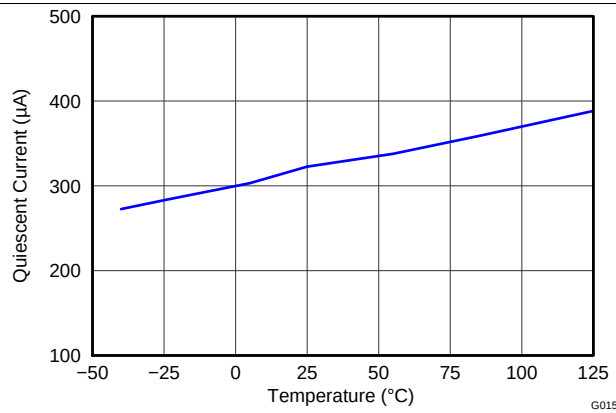


图 15. Active I_Q vs Temperature

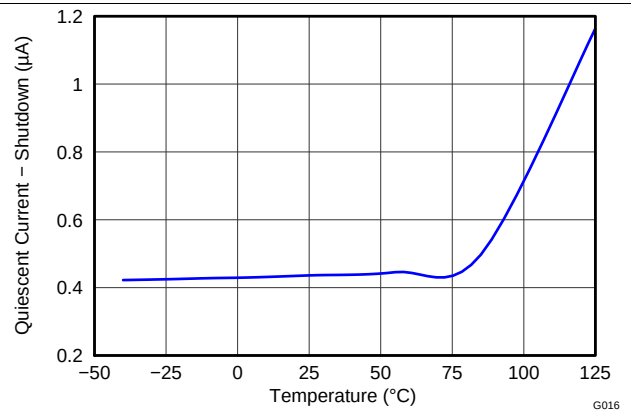


图 16. Shutdown I_Q vs Temperature

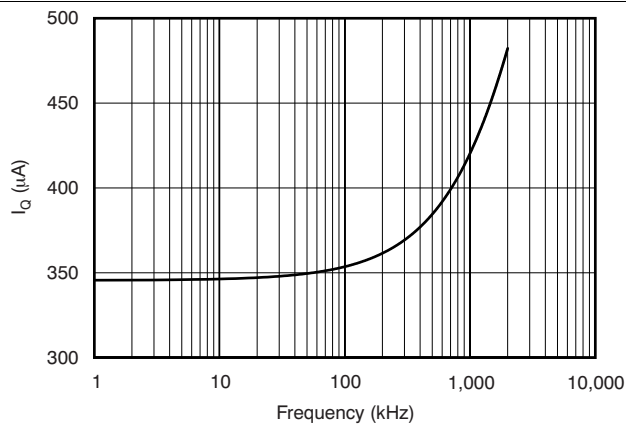


图 17. Active I_Q vs I^2C Clock Frequency

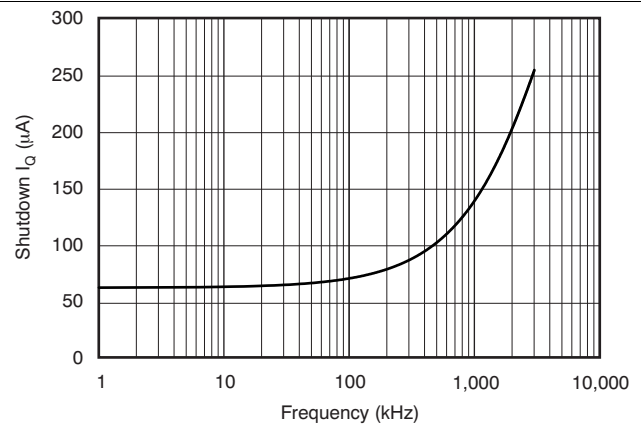


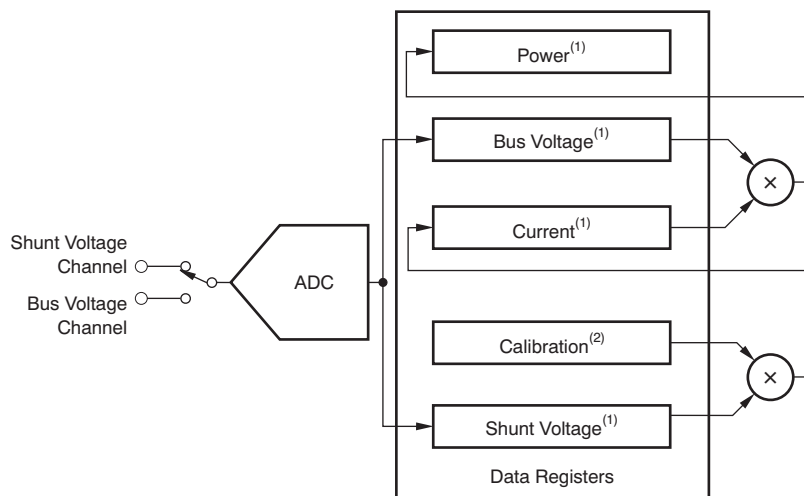
图 18. Shutdown I_Q vs I^2C Clock Frequency

7 Detailed Description

7.1 Overview

The INA226 is a digital current sense amplifier with an I²C- and SMBus-compatible interface. It provides digital current, voltage, and power readings necessary for accurate decision-making in precisely-controlled systems. Programmable registers allow flexible configuration for measurement resolution as well as continuous-versus-triggered operation. Detailed register information appears at the end of this data sheet, beginning with 表 4. See the [Functional Block Diagram](#) section for a block diagram of the INA226 device.

7.2 Functional Block Diagram



(1) Read-only

(2) Read/write

7.3 Feature Description

7.3.1 Basic ADC Functions

The INA226 device performs two measurements on the power-supply bus of interest. The voltage developed from the load current that flows through a shunt resistor creates a shunt voltage that is measured at the IN+ and IN– pins. The device can also measure the power supply bus voltage by connecting this voltage to the VBUS pin. The differential shunt voltage is measured with respect to the IN– pin while the bus voltage is measured with respect to ground.

The device is typically powered by a separate supply that can range from 2.7 V to 5.5 V. The bus that is being monitored can range in voltage from 0 V to 36 V. Based on the fixed 1.25-mV LSB for the Bus Voltage Register that a full-scale register results in a 40.96 V value.

注

Do not apply more than 36 V of actual voltage to the input pins.

There are no special considerations for power-supply sequencing because the common-mode input range and power-supply voltage are independent of each other; therefore, the bus voltage can be present with the supply voltage off, and vice-versa.

The device takes two measurements, shunt voltage and bus voltage. It then converts these measurements to current, based on the Calibration Register value, and then calculates power. Refer to the [Programming the Calibration Register](#) section for additional information on programming the Calibration Register.

Feature Description (接下页)

The device has two operating modes, continuous and triggered, that determine how the ADC operates following these conversions. When the device is in the normal operating mode (that is, MODE bits of the Configuration Register (00h) are set to '111'), it continuously converts a shunt voltage reading followed by a bus voltage reading. After the shunt voltage reading, the current value is calculated (based on 公式 3). This current value is then used to calculate the power result (using 公式 4). These values are subsequently stored in an accumulator, and the measurement/calculation sequence repeats until the number of averages set in the Configuration Register (00h) is reached. Following every sequence, the present set of values measured and calculated are appended to previously collected values. After all of the averaging has been completed, the final values for shunt voltage, bus voltage, current, and power are updated in the corresponding registers that can then be read. These values remain in the data output registers until they are replaced by the next fully completed conversion results. Reading the data output registers does not affect a conversion in progress.

The mode control in the Conversion Register (00h) also permits selecting modes to convert only the shunt voltage or the bus voltage in order to further allow the user to configure the monitoring function to fit the specific application requirements.

All current and power calculations are performed in the background and do not contribute to conversion time.

In triggered mode, writing any of the triggered convert modes into the Configuration Register (00h) (that is, MODE bits of the Configuration Register (00h) are set to '001', '010', or '011') triggers a single-shot conversion. This action produces a single set of measurements; thus, to trigger another single-shot conversion, the Configuration Register (00h) must be written to a second time, even if the mode does not change.

In addition to the two operating modes (continuous and triggered), the device also has a power-down mode that reduces the quiescent current and turns off current into the device inputs, reducing the impact of supply drain when the device is not being used. Full recovery from power-down mode requires 40μs. The registers of the device can be written to and read from while the device is in power-down mode. The device remains in power-down mode until one of the active modes settings are written into the Configuration Register (00h) .

Although the device can be read at any time, and the data from the last conversion remain available, the Conversion Ready flag bit (Mask/Enable Register, CVRF bit) is provided to help coordinate one-shot or triggered conversions. The Conversion Ready flag (CVRF) bit is set after all conversions, averaging, and multiplication operations are complete.

The Conversion Ready flag (CVRF) bit clears under these conditions:

- Writing to the Configuration Register (00h), except when configuring the MODE bits for power-down mode; or
- Reading the Mask/Enable Register (06h)

7.3.1.1 Power Calculation

The Current and Power are calculated following shunt voltage and bus voltage measurements as shown in 图 19. Current is calculated following a shunt voltage measurement based on the value set in the Calibration Register. If there is no value loaded into the Calibration Register, the current value stored is zero. Power is calculated following the bus voltage measurement based on the previous current calculation and bus voltage measurement. If there is no value loaded in the Calibration Register, the power value stored is also zero. Again, these calculations are performed in the background and do not add to the overall conversion time. These current and power values are considered intermediate results (unless the averaging is set to 1) and are stored in an internal accumulation register, not the corresponding output registers. Following every measured sample, the newly-calculated values for current and power are appended to this accumulation register until all of the samples have been measured and averaged based on the number of averages set in the Configuration Register (00h).

Feature Description (接下页)

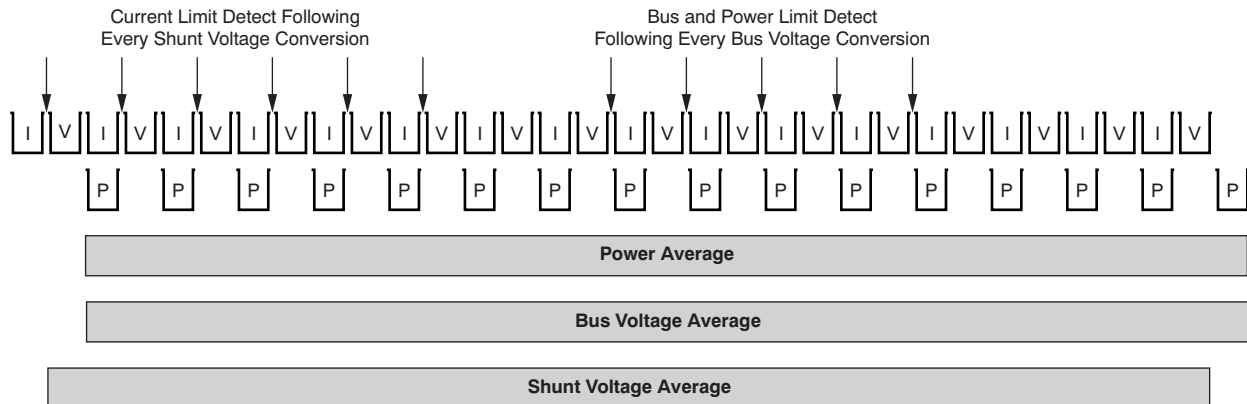


图 19. Power Calculation Scheme

除了在每次采样后累加电流和功率值之外，分流电压和总线电压的测量值也会被收集。在所有采样都完成并且进行了相应的电流和功率计算之后，每个参数的累积平均值会被加载到相应的输出寄存器中，然后就可以从这些寄存器中读取这些平均值了。

7.3.1.2 Alert Pin

INA226 具有一个单独的报警限制寄存器 (07h)，该寄存器允许将报警 (Alert) 引脚编程为响应用户自定义的单个事件，或者根据需要响应转换就绪通知。屏蔽/使能寄存器允许用户从五个可用的功能中选择一个进行监控，并且/或者设置转换就绪位来控制报警引脚的响应。

根据正在监控的功能，用户需要在报警限制寄存器中输入一个数值，以设置触发报警引脚的相应阈值。

报警 (Alert) 引脚允许监控几个可用的报警功能中的一个，以确定是否已超出用户定义的阈值。可以监控的五个报警功能是：

- 1) 分流器过压限制 SOL
- 2) 分流器欠压限制 SUL
- 3) 总线过压限制 BOL
- 4) 总线欠压限制 BUL
- 5) 功率超限 POL

报警 (Alert) 引脚是一个开漏输出。当屏蔽/使能寄存器中选择的报警功能超出了报警限制寄存器中编程的值时，该引脚将被置为有效状态。在同一时间，只能启用和监控这些报警功能中的一个。如果启用了多个报警功能，则最高有效位位置上选择的功能具有更高的优先级，并根据报警限制寄存器的值做出响应。例如，如果同时选择了分流器过压限制功能和分流器欠压限制功能，那么当分流器电压寄存器的值超过报警限制寄存器的值时，报警引脚将被置为有效状态。

器件的转换就绪状态也可以在报警 (Alert) 引脚上进行监控，以通知用户器件何时完成了之前的转换并准备好开始新的转换。

转换就绪状态可以与一个报警功能同时在报警引脚上进行监控。如果同时启用了某个报警功能和转换就绪状态在报警引脚上进行监控，那么在报警引脚被置为有效状态后，必须读取屏蔽/使能寄存器才能确定报警的来源。

通过读取屏蔽/使能寄存器中的转换就绪标志 (CVRF, bit 3) 和报警功能标志 (AFF, bit 4)，可以确定报警的来源。

如果不需要转换就绪功能且未设置 CNVR 位，则报警引脚仅响应基于启用的报警功能而发生的超出报警限制的情况。

如果未使用报警功能，报警 (Alert) 引脚可以悬空，而不会影响器件的运行。

Feature Description (接下页)

Refer to 图 19 to see the relative timing of when the value in the Alert Limit Register is compared to the corresponding converted value. For example, if the alert function that is enabled is Shunt Voltage Over-Limit (SOL), following every shunt voltage conversion the value in the Alert Limit Register is compared to the measured shunt voltage to determine if the measurements has exceeded the programmed limit. The AFF, bit 4 of the Mask/Enable Register, asserts high any time the measured voltage exceeds the value programmed into the Alert Limit Register. In addition to the AFF being asserted, the Alert pin is asserted based on the Alert Polarity Bit (APOL, bit 1 of the Mask/Enable Register). If the Alert Latch is enabled, the AFF and Alert pin remain asserted until either the Configuration Register (00h) is written to or the Mask/Enable Register is read.

The Bus Voltage alert functions compare the measured bus voltage to the Alert Limit Register following every bus voltage conversion and assert the AFF bit and Alert pin if the limit threshold is exceeded.

The Power Over-Limit alert function is also compared to the calculated power value following every bus voltage measurement conversion and asserts the AFF bit and Alert pin if the limit threshold is exceeded.

7.4 Device Functional Modes

7.4.1 Averaging and Conversion Time Considerations

The INA226 device offers programmable conversion times (t_{CT}) for both the shunt voltage and bus voltage measurements. The conversion times for these measurements can be selected from as fast as 140 μ s to as long as 8.244 ms. The conversion time settings, along with the programmable averaging mode, allow the device to be configured to optimize the available timing requirements in a given application. For example, if a system requires that data be read every 5ms, the device could be configured with the conversion times set to 588 μ s for both shunt and bus voltage measurements and the averaging mode set to 4. This configuration results in the data updating approximately every 4.7ms. The device could also be configured with a different conversion time setting for the shunt and bus voltage measurements. This type of approach is common in applications where the bus voltage tends to be relatively stable. This situation can allow for the time focused on the bus voltage measurement to be reduced relative to the shunt voltage measurement. The shunt voltage conversion time could be set to 4.156 ms with the bus voltage conversion time set to 588 μ s, with the averaging mode set to 1. This configuration also results in data updating approximately every 4.7 ms.

There are trade-offs associated with the settings for conversion time and the averaging mode used. The averaging feature can significantly improve the measurement accuracy by effectively filtering the signal. This approach allows the device to reduce any noise in the measurement that may be caused by noise coupling into the signal. A greater number of averages enables the device to be more effective in reducing the noise component of the measurement.

The conversion times selected can also have an impact on the measurement accuracy. 图 20 shows multiple conversion times to illustrate the impact of noise on the measurement. In order to achieve the highest accuracy measurement possible, use a combination of the longest allowable conversion times and highest number of averages, based on the timing requirements of the system.

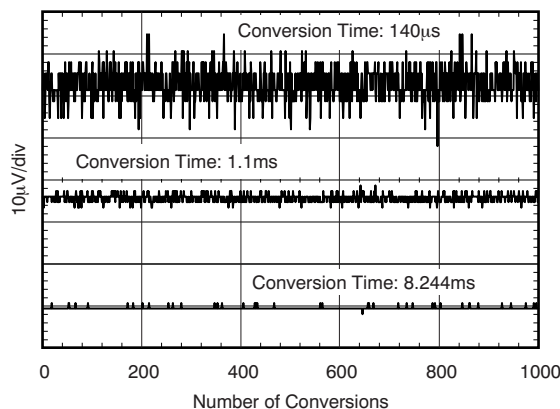


图 20. Noise vs Conversion Time

7.5 Programming

INA226 器件的一个重要方面是，它本身并不直接测量电流或功率。该器件测量的是施加在 IN+ 和 IN- 输入引脚之间的差分电压，以及施加在 VBUS 引脚上的电压。

为了让器件报告电流和功率值，用户必须编程电流寄存器(04h)的分辨率以及应用中使用的分流电阻的阻值，该分流电阻用于产生施加在输入引脚之间的差分电压。功率寄存器(03h) 在内部被设定为电流 LSB 编程值的 25 倍。

电流 LSB 和分流电阻值都用于计算校准寄存器的值，器件会使用该校准寄存器的值，基于测得的分流电压和总线电压来计算相应的电流和功率值

校准寄存器的计算基于公式 1。

该公式包含电流 LSB (Current_LSB) 项，它是为电流寄存器 (04h) 编程的最低有效位的值。

用户使用这个值将电流寄存器 (04h) 中的数值转换为实际电流值（单位为安培）。

电流寄存器 (04h) 的最高分辨率可以通过使用基于最大预期电流的最小允许电流 LSB 来获得，如公式 2 所示。

虽然这个值可以提供最高的分辨率，但通常会选择一个略大于该值的、最接近的整数作为电流 LSB，以简化电流寄存器 (04h) 和功率寄存器(03h) 到安培和瓦特的转换。

R SHUNT 项是用于在输入引脚之间产生差分电压的外部shunt电阻的阻值

$$CAL = \frac{0.00512}{Current_LSB \cdot R_{SHUNT}}$$

where

•0.00512 是一个内部固定值，用于确保比例关系得到正确维持 (1)

$$Current_LSB = \frac{Maximum\ Expected\ Current}{2^{15}} \quad (2)$$

在编程校准寄存器之后，电流寄存器 (04h) 和功率寄存器 (03h) 会根据相应的分流电压和总线电压测量值进行更新。

在校准寄存器被编程之前，电流寄存器 (04h) 和功率寄存器 (03h) 的值将保持为0

7.5.1 Programming the Calibration Register

图 27 这里展示了一个标称 10 A 的负载，它在一个 2 mΩ 的分流电阻上产生了 20 mV 的差分电压。INA226 的总线电压是在外部 VBUS 输入引脚上测量的，在本例中，该引脚连接到 IN- 引脚，以测量输送到负载的电压电平。在这个例子中，VBUS 引脚测得的电压小于 12 V，因为 IN- 引脚上的电压是 11.98 V，这是由于分流电阻上的电压降造成的。

在这个例子中，假设最大预期电流为 15 A，使用公式 2 计算出的电流 LSB (Current_LSB) 为 457.7 μA/bit。如果使用 500 μA/bit 或 1 mA/bit 的电流 LSB 值，将大大简化电流寄存器 (04h) 和功率寄存器 (03h) 到安培和瓦特的转换。在本例中，选择了 1 mA/bit 的电流 LSB 值。使用这个电流 LSB 值确实牺牲了一小部分分辨率，但换来了用户端更简单的转换过程。在本例中，使用公式 1，电流 LSB 值为 1 mA/bit，分流电阻为 2 mΩ，计算出的校准寄存器值为 2560，即 A00h。

然后，电流寄存器 (04h) 的值通过将分流器电压寄存器 (01h) 中的十进制数值乘以校准寄存器的十进制数值再除以 2048 来计算，如公式 3 所示。在本例中，分流器电压寄存器包含的值为 8000（代表 20 mV）

将其乘以校准寄存器的值 2560，然后除以 2048，得到电流寄存器 (04h) 的十进制数值 10000，即 2710h 将这个值乘以 1 mA/bit 的电流 LSB，即可得到示例中提到的原始 10 A 电流值。

$$Current = \frac{ShuntVoltage \cdot CalibrationRegister}{2048} \quad (3)$$

Programming (接下页)

总线电压寄存器 (02h) 的 LSB 是一个固定的 1.25 mV/bit，这意味着 VBUS 引脚上 11.98 V 的电压值将对应于寄存器值 2570h，或十进制的 9584。请注意，总线电压寄存器 (02h) 的最高有效位 (MSB) 始终为零，因为 VBUS 引脚只能测量正电压。

功率寄存器 (03h) 的值通过将电流寄存器的十进制数值 10000 乘以总线电压寄存器 (02h) 的十进制数值 9584，再除以 20000 来计算，如公式 4 所定义。

在本例中，功率寄存器 (03h) 的结果是 12B8h，或十进制的 4792。将这个结果乘以功率 LSB (25 倍的 $[1 \times 10^{-3} \text{ Current_LSB}]$)，得到功率计算值为 $(4792 \times 25 \text{ mW/bit})$ ，即 119.8 W。

功率 LSB 与电流 LSB 的比率固定为 25。在本例中，编程的 1 mA/bit 电流 LSB 导致功率 LSB 为 25 mW/bit。

这个比率在内部被预先设定，以确保功率计算的比例在可接受的范围内。手动计算输送到负载的功率，可以使用 11.98 V 的总线电压 (12 V 共模电压 – 20 mV 分流压降) 乘以 10 A 的负载电流，得到 119.8 W 的结果。

$$\text{Power} = \frac{\text{Current} \cdot \text{BusVoltage}}{20,000} \quad (4)$$

表 1 列出了配置、测量和计算该器件的电流和功率值的步骤。

表 1. 电流和功率计算⁽¹⁾

STEP	REGISTER NAME	ADDRESS	CONTENTS	DEC	LSB	VALUE
Step 1	Configuration Register	00h	4127h	—	—	—
Step 2	Shunt Register	01h	1F40h	8000	2.5 μV	20 mV
Step 3	Bus Voltage Register	02h	2570h	9584	1.25 mV	11.98 V
Step 4	Calibration Register	05h	A00h	2560	—	—
Step 5	Current Register	04h	2710	10000	1 mA	10 A
Step 6	Power Register	03h	12B8h	4792	25 mW	119.82 W

(1) Conditions: Load = 10 A, $V_{\text{CM}} = 12 \text{ V}$, $R_{\text{SHUNT}} = 2 \text{ m}\Omega$, and $V_{\text{VBUS}} = 12 \text{ V}$.

7.5.2 功率测量引擎编程

7.5.2.1 校准寄存器和比例换算

校准寄存器使得我们能够将电流寄存器 (04h) 和功率寄存器 (03h) 的数值缩放到对于给定应用最有用的范围。例如，可以设置校准寄存器，使得在预期的满量程点，电流寄存器 (04h) 或功率寄存器 (03h) 中产生尽可能大的数值。这种方法利用之前计算出的校准寄存器公式中的最小电流 LSB，从而获得最高的分辨率。

也可以选择校准寄存器，使得电流寄存器 (04h) 和功率寄存器 (03h) 中的数值直接对应被测量值的十进制等效值，或者为每个相应的寄存器产生一个近似的 LSB 值。在做出这些选择之后，校准寄存器还为最终用户的系统级校准提供了可能性。

通过使用外部电流表确定精确的电流值后，可以根据 INA226 的测量电流结果调整校准寄存器的值，以消除整个系统的误差，如 公式 5 所示

$$\text{Corrected_Full_Scale_Cal} = \text{trunc} \left[\frac{\text{Cal} \times \text{MeasShuntCurrent}}{\text{INA226_Current}} \right] \quad (5)$$

7.5.3 简单的电流分流监视器用法 (无需编程)

如果只需要读取分流器电压降和总线电压，并且使用默认的上电复位配置以及连续转换分流器和总线电压的模式，则该器件无需任何编程即可使用

如果不编程器件的校准寄存器，该器件将无法提供有效的电流或功率值，因为这两个输出都是使用加载到校准寄存器中的值导出的

7.5.4 Default Settings

The default power-up states of the registers are shown in the [Register Maps](#) section of this data sheet. These registers are volatile, and if programmed to a value other than the default values shown in [表 4](#), they must be re-programmed at every device power-up. Detailed information on programming the Calibration Register specifically is given in the [Programming](#) section and calculated based on [公式 1](#).

7.5.4默认设置寄存器的默认上电状态在本数据手册的寄存器映射部分中显示。这些寄存器是易失性的，如果被编程为一个值，而不是在4中显示的默认值，它们必须在每次设备上电时重新编程。关于校准寄存器编程的详细信息在编程一节中给出，并根据通用表格1进行计算。

7.5.5 Bus Overview

INA226 兼容 I²C 和 SMBus 接口。I²C 和 SMBus 协议本质上是相互兼容的。

在本数据手册中，I²C 接口被用作主要示例，只有在讨论两种系统之间的差异时才具体说明 SMBus 协议。两条线路，SCL 和 SDA，将器件连接到总线。SCL 和 SDA 都是开漏连接

发起数据传输的器件称为主器件，受主器件控制的器件是从器件。总线必须由一个主器件控制，该主器件产生串行时钟 (SCL)，控制总线访问，并产生起始 (START) 和停止 (STOP) 条件。

为了寻址特定的器件，主器件通过在 SCL 为高电平时将数据信号线 (SDA) 从高电平拉低来启动一个起始条件。总线上的所有从器件在 SCL 的上升沿移入从器件地址字节，最后一个比特指示是读操作还是写操作。在第九个时钟脉冲期间，被寻址的从器件通过产生一个应答 (Acknowledge) 并将 SDA 拉低来响应主器件。

然后启动数据传输，每次发送八位数据，后跟一个应答位。在数据传输期间，当 SCL 为高电平时，SDA 必须保持稳定。当 SCL 为高电平时，SDA 上的任何变化都被解释为起始或停止条件。

所有数据传输完毕后，主器件产生一个停止条件，通过在 SCL 为高电平时将 SDA 从低电平拉高来指示。该器件在其接口上包含一个 28 ms 的超时，以防止总线锁定。

7.5.5.1 Serial Bus Address

要与 INA226 通信，主器件必须首先通过从器件地址字节寻址从器件。从器件地址字节由七位地址位和一个方向位组成，该方向位指示操作是读取还是写入。

该器件有两个地址引脚，A0 和 A1。表 2 列出了 16 个可能的地址中每个地址对应的引脚逻辑电平。器件在每次总线通信时都会采样 A0 和 A1 引脚的状态。在接口上发生任何活动之前，请先确定这些引脚的状态。

表 2. Address Pins and Slave Addresses

A1	A0	SLAVE ADDRESS
GND	GND	1000000
GND	VS	1000001
GND	SDA	1000010
GND	SCL	1000011
VS	GND	1000100
VS	VS	1000101
VS	SDA	1000110
VS	SCL	1000111
SDA	GND	1001000
SDA	VS	1001001
SDA	SDA	1001010
SDA	SCL	1001011
SCL	GND	1001100
SCL	VS	1001101
SCL	SDA	1001110
SCL	SCL	1001111

7.5.5.2 Serial Interface

INA226 在 I²C 总线和 SMBus 总线上都只作为从器件运行。与总线的连接通过开漏的 SDA 和 SCL 线路进行。SDA 和 SCL 引脚集成了尖峰抑制滤波器和施密特触发器，以最大限度地减少输入尖峰和总线噪声的影响。虽然该器件在数字 I/O 线路中集成了尖峰抑制功能，但适当的布局技术有助于最大限度地减少耦合到通信线路中的噪声量。这种噪声引入可能来自两条通信线路自身之间的容性耦合信号边沿，或者来自系统中存在的其他开关噪声源。在印刷电路板 (PCB) 的不同层上，将走线与中间的接地层并行布线，通常可以减少通信线路之间的耦合效应。屏蔽通信线路可以降低意外噪声耦合到数字 I/O 线路的可能性，这些噪声可能会被错误地解释为起始或停止命令。

INA226 支持快速模式（1 kHz 至 400 kHz）和高速模式（1 kHz 至 2.94 MHz）的传输协议。所有数据字节都以最高有效位优先 (MSB first) 的顺序进行传输

7.5.5.3 Writing to and Reading from the INA226

Accessing a specific register on the INA226 is accomplished by writing the appropriate value to the register pointer. Refer to 表 4 for a complete list of registers and corresponding addresses. The value for the register pointer (as shown in 图 25) is the first byte transferred after the slave address byte with the R/W bit low. Every write operation to the device requires a value for the register pointer.

向寄存器写入数据始于主器件发送的第一个字节。这个字节是从器件地址，R/W 位为低电平。然后，器件应答接收到一个有效的地址。

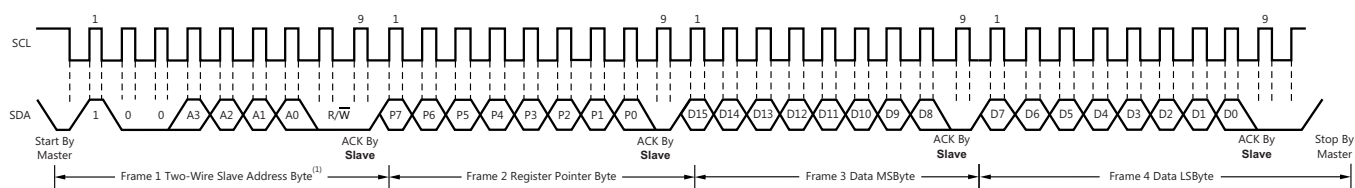
主器件发送的下一个字节是要写入数据的寄存器的地址。这个寄存器地址值将寄存器指针更新到所需的寄存器。接下来的两个字节被写入由寄存器指针寻址的寄存器。器件应答接收到每个数据字节。主器件可以通过产生一个起始或停止条件来终止数据传输。

当从器件读取数据时，上一次写入操作存储在寄存器指针中的值决定了在读取操作期间读取哪个寄存器。要更改读取操作的寄存器指针，必须向寄存器指针写入一个新值。此写入通过发送从器件地址字节（R/W 位为低电平），后跟寄存器指针字节来完成。不需要额外的写入数据。然后，主器件产生一个起始条件，并发送从器件地址字节（R/W 位为高电平）以启动读取命令。下一个字节由从器件发送，并且是由寄存器指针指示的寄存器的最高有效字节 (MSB)。主器件发送一个应答 (Acknowledge)；然后，从器件发送最低有效字节 (LSB)。主器件应答接收到该数据字节。主器件可以通过在接收到任何数据字节后产生一个非应答 (Not-Acknowledge)，或者产生一个起始或停止条件来终止数据传输。如果需从同一个寄存器重复读取，则不必持续发送寄存器指针字节；器件会保留寄存器指针的值，直到下一次写入操作更改它。

图 22 shows the write operation timing diagram. 图 23 shows the read operation timing diagram.

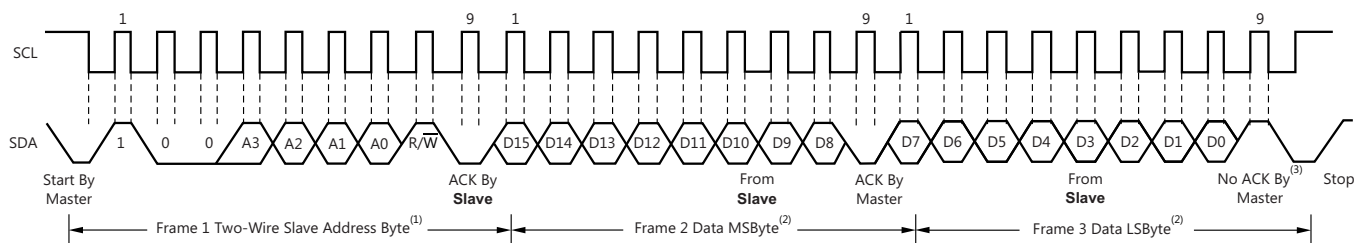
注

Register bytes are sent most-significant byte first, followed by the least significant byte.



- (1) The value of the Slave Address byte is determined by the settings of the A0 and A1 pins. Refer to 表 2.

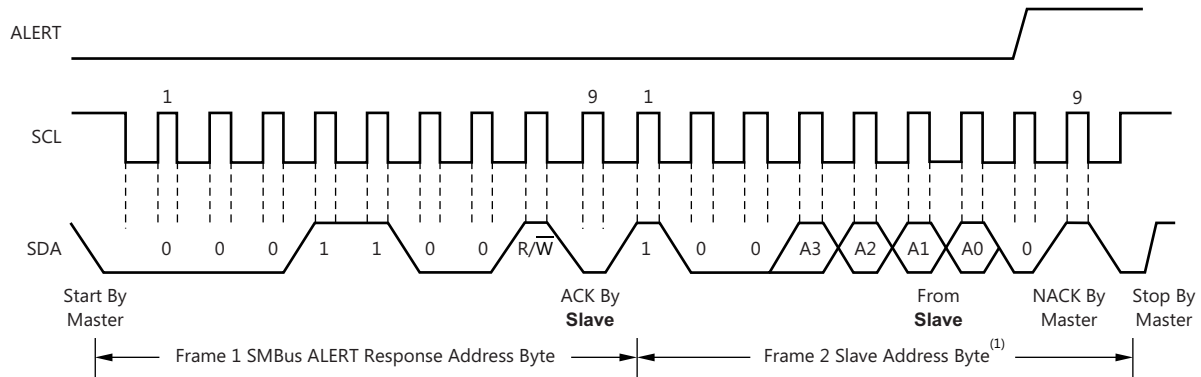
图 22. Timing Diagram for Write Word Format



- (1) The value of the Slave Address byte is determined by the settings of the A0 and A1 pins. Refer to 表 2.
(2) Read data is from the last register pointer location. If a new register is desired, the register pointer must be updated. See 图 25.
(3) ACK by Master can also be sent.

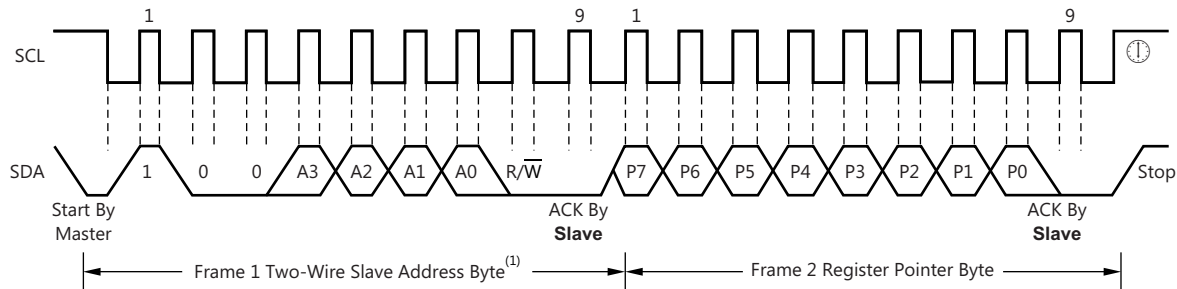
图 23. Timing Diagram for Read Word Format

图 24 shows the timing diagram for the SMBus Alert response operation. 图 25 illustrates a typical register pointer configuration.



(1) The value of the Slave Address Byte is determined by the settings of the A0 and A1 pins. Refer to [表 2](#).

图 24. Timing Diagram for SMBus ALERT



(1) The value of the Slave Address Byte is determined by the settings of the A0 and A1 pins. Refer to [表 2](#).

图 25. Typical Register Pointer Set

7.5.5.3.1 High-Speed I²C Mode

当总线空闲时，SDA 和 SCL 线都被上拉电阻拉至高电平。主器件产生一个起始条件，后跟一个包含高速 (HS) 主代码 00001XXX 的有效串行字节。

此传输以快速模式 (400 kHz) 或标准模式 (100 kHz) (F/S) 进行，速度不超过 400 kHz。INA226 不会应答 HS 主代码，但会识别它并切换其内部滤波器以支持 2.94 MHz 的操作。

主器件产生一个重复起始条件（重复起始条件与起始条件具有相同的时序）。

在这个重复起始条件之后，协议与快速/标准 (F/S) 模式相同，只是允许高达 2.94 MHz 的传输速度。

在 HS 模式下，不要使用停止条件，而是使用重复起始条件来保持总线占用。停止条件会结束 HS 模式，并将器件的所有内部滤波器切换回支持 F/S 模式。

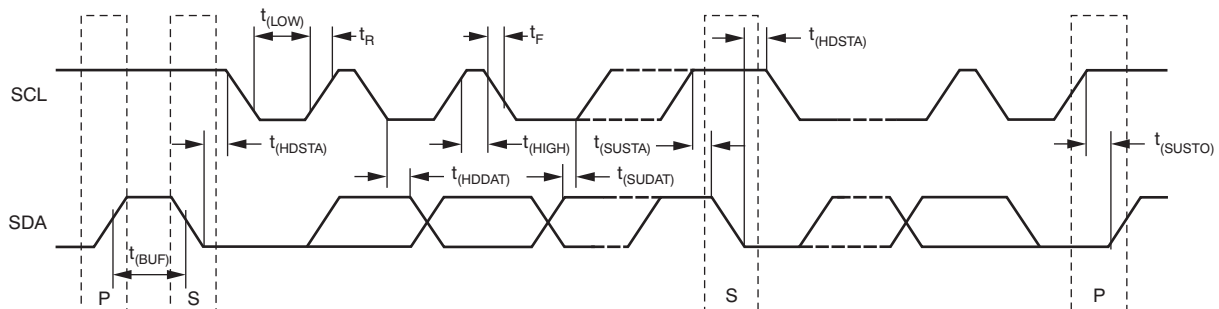


图 26. Bus Timing Diagram

表 3. Bus Timing Diagram Definitions⁽¹⁾

PARAMETER		FAST MODE		HIGH-SPEED MODE		UNIT
		MIN	MAX	MIN	MAX	
SCL operating frequency	$f_{(SCL)}$	0.001	0.4	0.001	2.94	MHz
Bus free time between stop and start conditions	$t_{(BUF)}$	600		160		ns
Hold time after repeated START condition. After this period, the first clock is generated.	$t_{(HDSTA)}$	100		100		ns
Repeated start condition setup time	$t_{(SUSTA)}$	100		100		ns
STOP condition setup time	$t_{(SUSTO)}$	100		100		ns
Data hold time	$t_{(HDDAT)}$	10	900	10	100	ns
Data setup time	$t_{(SUDAT)}$	100		20		ns
SCL clock low period	$t_{(LOW)}$	1300		200		ns
SCL clock high period	$t_{(HIGH)}$	600		60		ns
Data fall time	t_F		300		80	ns
Clock fall time	t_F		300		40	ns
Clock rise time	t_R		300		40	ns
Clock/data rise time for $SCLK \leq 100kHz$	t_R		1000			ns

(1) Values based on a statistical analysis of a one-time sample of devices. Minimum and maximum values are not guaranteed and not production tested.

7.5.5.4 SMBus Alert Response SMBus 报警响应

INA226 的设计使其能够响应 SMBus 的报警响应地址。对于简单的从器件来说，SMBus 报警响应提供了一种快速识别故障的方法。当有报警发生时，主器件可以发送一个广播，目标地址是报警响应从地址 (0001100)，并且设置读/写位为读取状态。收到这个广播后，任何产生了报警的从器件都会通过发送一个应答信号，并在总线上广播自己的地址来表明身份。

报警响应可以同时激活多个从器件，这与 I²C 的通用呼叫类似。如果有多个从器件尝试响应，则适用总线仲裁规则。仲裁失败的器件不会产生应答信号，并继续将报警线路拉低，直到中断被清除。

7.6 Register Maps

The INA226 uses a bank of registers for holding configuration settings, measurement results, minimum/maximum limits, and status information. 表 4 summarizes the device registers; refer to the [Functional Block Diagram](#) section for an illustration of the registers.

All 16-bit device registers are two 8-bit bytes via the I²C interface.

表 4. Register Set Summary

POINTER ADDRESS	REGISTER NAME	FUNCTION	POWER-ON RESET		TYPE ⁽¹⁾
			BINARY	HEX	
00h	Configuration Register	All-register reset, shunt voltage and bus voltage ADC conversion times and averaging, operating mode.	01000001 00100111	4127	R/W
01h	Shunt Voltage Register	Shunt voltage measurement data.	00000000 00000000	0000	R
02h	Bus Voltage Register	Bus voltage measurement data.	00000000 00000000	0000	R
03h	Power Register ⁽²⁾	Contains the value of the calculated power being delivered to the load.	00000000 00000000	0000	R
04h	Current Register ⁽²⁾	Contains the value of the calculated current flowing through the shunt resistor.	00000000 00000000	0000	R

(1) Type: R = Read-Only, R/W = Read/Write.

(2) The Current Register (04h) and Power Register (03h) default to '0' because the Calibration register defaults to '0', yielding zero current and power values until the Calibration register is programmed.

Register Maps (接下页)

表 4. Register Set Summary (接下页)

POINTER ADDRESS	REGISTER NAME	FUNCTION	POWER-ON RESET		TYPE ⁽¹⁾
			BINARY	HEX	
05h	Calibration Register	设置电流和功率测量的满量程范围和 LSB。整体系统校准	00000000 00000000	0000	R/ $\overline{W}$
06h	Mask/Enable Register	报警配置和转换就绪标志。	00000000 00000000	0000	R/ $\overline{W}$
07h	Alert Limit Register	包含与所选报警功能进行比较的限值。	00000000 00000000	0000	R/ $\overline{W}$
FEh	Manufacturer ID Register	包含唯一的制造商识别号。	0101010001001001	5449	R
FFh	Die ID Register	包含唯一的芯片识别号	0010001001100000	2260	R

7.6.1 Configuration Register (00h) (Read/Write)

表 5. Configuration Register (00h) (Read/Write) Descriptions

BIT NO.	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	RST	—	—	—	AVG2	AVG1	AVG0	VBUSCT2	VBUSCT1	VBUSCT0	VSHCT2	VSHCT1	VSHCT0	MODE3	MODE2	MODE1
POR VALUE	0	1	0	0	0	0	0	1	0	0	1	0	0	1	1	1

配置寄存器的设置用以控制器件的工作模式。该寄存器用以控制分流电压¹和总线电压测量的转换时间设置，以及所采用的平均模式。用于控制选择测量哪些信号的工作模式，亦在配置寄存器中编程设定。配置寄存器可随时读取，而不会影响器件设置或正在进行的转换。对配置寄存器执行写操作，会中止任何正在进行的转换，直至写操作序列完成；此后，将基于配置寄存器 (00h) 的新内容开始一次新的转换。此中止操作可避免下一次完成的转换在其所用条件上出现任何不确定性

RST:	Reset Bit
Bit 15	将该位设置为 '1' 会触发一次系统复位，其效果等同于上电复位。此操作会将所有寄存器复位至默认值；该（设置）位会自动清零
AVG:	Averaging Mode 求平均值 模式
Bits 9–11	（此设置/这些位）用于确定所采集并进行平均的采样数量。表6列出了所有AVG ¹ 位的设置以及每种设置下对应的平均采样次数。

表 6. AVG Bit Settings[11:9] Combinations

AVG2 D11	AVG1 D10	AVG0 D9	NUMBER OF AVERAGES ⁽¹⁾
0	0	0	1
0	0	1	4
0	1	0	16
0	1	1	64
1	0	0	128
1	0	1	256
1	1	0	512
1	1	1	1024

(1) Shaded values are default.

VBUSCT:

Bits 6–8

Bus Voltage Conversion Time 总线电压转换时间

Sets the conversion time for the bus voltage measurement. 表 7 shows the VBUSCT bit options and related conversion times for each bit setting.

表 7. VBUSCT Bit Settings [8:6] Combinations

VBUSCT2 D8	VBUSCT1 D7	VBUSCT0 D6	CONVERSION TIME ⁽¹⁾
0	0	0	140 μ s
0	0	1	204 μ s
0	1	0	332 μ s
0	1	1	588 μ s
1	0	0	1.1 ms
1	0	1	2.116 ms
1	1	0	4.156 ms
1	1	1	8.244 ms

(1) Shaded values are default.

VSHCT:

Bits 3–5

Shunt Voltage Conversion Time 分流电压转换时间

Sets the conversion time for the shunt voltage measurement. 表 8 shows the VSHCT bit options and related conversion times for each bit setting.

表 8. VSHCT Bit Settings [5:3] Combinations

VSHCT2 D8	VSHCT1 D7	VSHCT0 D6	CONVERSION TIME ⁽¹⁾
0	0	0	140 μ s
0	0	1	204 μ s
0	1	0	332 μ s
0	1	1	588 μ s
1	0	0	1.1 ms
1	0	1	2.116 ms
1	1	0	4.156 ms
1	1	1	8.244 ms

(1) Shaded values are default.

MODE:

Bits 0-2

Operating Mode 模式选择

选择连续、触发或掉电工作模式。这些位默认为连续分流器和总线测量模式。模式设置如 表 9.

表 9. Mode Settings [2:0] Combinations

MODE3 D2	MODE2 D1	MODE1 D0	MODE ⁽¹⁾
0	0	0	Power-Down (or Shutdown)
0	0	1	Shunt Voltage, Triggered
0	1	0	Bus Voltage, Triggered
0	1	1	Shunt and Bus, Triggered
1	0	0	Power-Down (or Shutdown)
1	0	1	Shunt Voltage, Continuous
1	1	0	Bus Voltage, Continuous
1	1	1	Shunt and Bus, Continuous

(1) Shaded values are default.

7.6.2 分流器电压寄存器 (01h) (只读)

分流器电压寄存器存储当前分流器电压读数，V SHUNT。负数以二进制补码格式表示。生成负数的二进制补码的方法是：对该负数的绝对值的二进制数进行按位取反（求反），然后加 1。最高有效位 (MSB) 为 ‘1’ 表示负数。

示例：对于 V SHUNT = -80mV 的值：

1. 取绝对值：80mV
2. 将该数值转换为十进制整数： $(80\text{mV} \div 2.5\mu\text{V}) = 32000$
3. 将该数值转换为二进制：0111 1101 0000 0000
4. 对二进制结果进行按位取反：1000 0010 1111 1111
5. 对取反后的结果加 ‘1’ 以创建二进制补码结果： $1000 0010 1111 1111 + 1 = 1000 0011 0000 0000 = 8300\text{h}$

如果启用了平均功能，该寄存器将显示平均值。满量程范围 = 81.92mV (十进制 = 7FFFh)；最低有效位 (LSB) = 2.5μV。

表 10. Shunt Voltage Register (01h) (Read-Only) Description

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	SIGN	SD14	SD13	SD12	SD11	SD10	SD9	SD8	SD7	SD6	SD5	SD4	SD3	SD2	SD1	SD0
POR VALUE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

7.6.3 总线电压寄存器 (02h) (只读)

总线电压寄存器存储最新的总线电压读数，V BUS。如果启用了平均功能，该寄存器将显示平均值。满量程范围 = 40.96V (十进制 = 7FFFh)；最低有效位 (LSB) = 1.25mV。

表 11. Bus Voltage Register (02h) (Read-Only) Description

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	—	BD14	BD13	BD12	BD11	BD10	BD9	BD8	BD7	BD6	BD5	BD4	BD3	BD2	BD1	BD0
POR VALUE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

(1)(1) D15 始终为零，因为总线电压只能为正值。

7.6.4 功率寄存器 (03h) (只读)

如果启用了平均功能，该寄存器将显示平均值。

功率寄存器的最低有效位 (LSB) 在内部被设定为电流 LSB 设定值的 25 倍。功率寄存器用于记录功率值，单位为瓦特。其数值通过将电流寄存器的十进制数值与总线电压寄存器的十进制数值相乘计算得出，计算方法如下 [公式 4](#)。

4. 表 12. PowerRegister (03h) (Read-Only) Description

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	PD15	PD14	PD13	PD12	PD11	PD10	PD9	PD8	PD7	PD6	PD5	PD4	PD3	PD2	PD1	PD0
POR VALUE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

7.6.5 电流寄存器 (04h) (只读)

如果启用了平均功能，该寄存器将显示平均值。

电流寄存器中的数值是通过将分流器电压寄存器中的十进制数值与校准寄存器中的十进制数值相乘计算得出的，计算方法如下：[公式 3](#)。

表 13. Current Register (04h) (Read-Only) Register Description

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	CSIGN	CD14	CD13	CD12	CD11	CD10	CD9	CD8	CD7	CD6	CD5	CD4	CD3	CD2	CD1	CD0
POR VALUE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

7.6.6 校准寄存器 (05h) (读/写)

该寄存器为器件提供了用于产生所测差分电压的分流电阻值。它也设定了电流寄存器的分辨率。通过对该寄存器进行编程，可以设定电流 LSB (Current_LSB) 和功率 LSB (Power_LSB)。该寄存器也适用于整个系统的校准。关于校准寄存器编程的更多信息，请参见[Programming the Calibration Register](#)部分。

表 14. Calibration Register (05h) (Read/Write) Description

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	—	FS14	FS13	FS12	FS11	FS10	FS9	FS8	FS7	FS6	FS5	FS4	FS3	FS2	FS1	FS0
POR VALUE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

7.6.7 Mask/Enable Register (06h) (Read/Write)

屏蔽/使能寄存器用于选择启用哪些功能来控制报警 (Alert) 引脚的行为以及指示该引脚当前的功能。如果启用了多个功能，则最高有效位位置的报警功能 (Alert Function, D15-D11) 具有更高的优先级，并根据报警限制寄存器 (Alert Limit Register) 的设置做出响应。

表 15. Mask/Enable Register (06h) (Read/Write)

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	SOL	SUL	BOL	BUL	POL	CNVR	—	—	—	—	—	AFF	CVRF	OVF	APOL	LEN
POR VALUE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

SOL: Shunt Voltage Over-Voltage

Bit 15 将此位设置为高电平后，如果转换后的分流器电压测量值超过了报警限制寄存器中编程的值，报警 (Alert) 引脚将被置为有效状态。

SUL: Shunt Voltage Under-Voltage

Bit 14 将此位设置为高电平后，如果转换后的分流器电压测量值低于报警限制寄存器中编程的值，报警 (Alert) 引脚将被置为有效状态。

BOL: Bus Voltage Over-Voltage

Bit 13 将此位设置为高电平后，如果转换后的总线电压测量值超过了报警限制寄存器中编程的值，报警 (Alert) 引脚将被置为有效状态。

BUL: Bus Voltage Under-Voltage

Bit 12 将此位设置为高电平后，如果转换后的总线电压测量值低于报警限制寄存器中编程的值，报警 (Alert) 引脚将被置为有效状态。

POL: Power Over-Limit

Bit 11 将此位设置为高电平后，如果总线电压测量后计算出的功率值超过了报警限制寄存器中编程的值，报警 (Alert) 引脚将被置为有效状态。

CNVR: Conversion Ready

Bit 10 将此位设置为高电平后，当转换就绪标志 (Conversion Ready Flag)，即第 3 位被置为有效状态时，报警 (Alert) 引脚将被置为有效状态，表明器件已准备好进行下一次转换。

AFF: **Alert Function Flag** 报警功能标志

Bit 4 虽然在同一时间只有一个报警功能可以被报警引脚监控，但转换就绪标志也可以被使能来触发报警引脚。在报警发生后读取报警功能标志，可以帮助用户确定此次报警的来源是否是某个报警功能

当报警锁存使能位 (Alert Latch Enable bit) 设置为锁存模式时，报警功能标志位 (Alert Function Flag bit) 只有在屏蔽/使能寄存器被读取后才会清除。当报警锁存使能位设置为透明模式时，报警功能标志位在下次未导致报警的转换完成后就会清除

CVRF: **Conversion Ready Flag** 转换就绪标志

Bit 3

尽管可以随时读取器件的数据，并且上次转换的数据始终可用，但提供转换就绪标志位是为了帮助协调单次或触发式转换。在所有转换、平均和乘法运算完成后，转换就绪标志位将被置位。在以下情况下，转换就绪标志位会被清除：

- (1.) 写入配置寄存器 (除了选择掉电模式)
- (2.) 读取屏蔽/使能寄存器

OVF: **Math Overflow Flag** 算术溢出标志

Bit 2

如果算术运算导致溢出错误，则此位将被置为‘1’。它表明电流和功率数据可能无效

APOL: **Alert Polarity bit; sets the Alert pin polarity** 报警极性位；设置报警引脚的极性。

Bit 1

1 = 反相 (高电平有效，开漏输出)
0 = 正常 (低电平有效，开漏输出) (默认)

LEN: **Alert Latch Enable; configures the latching feature of the Alert pin and Alert Flag bits.**

Bit 0

1 = 锁存使能 即使报警条件消失，该标志位和报警引脚的输出状态也会保持不变需要手动清除
0 = 透明 (默认) 报警标志位和报警引脚的输出状态会随着报警条件的消失而自动恢复到非报警状态

当报警锁存使能位设置为透明模式时，一旦故障情况解除，报警引脚和标志位将自动复位到空闲状态。当报警锁存使能位设置为锁存模式时，在故障发生后，报警引脚和报警标志位将保持激活状态，直到屏蔽/使能寄存器被读取。

7.6.8 Alert Limit Register (07h) (Read/Write) 报警限制寄存器

报警限制寄存器 (Alert Limit Register) 包含一个数值，该数值将与屏蔽/使能寄存器中选定的寄存器的值进行比较，以确定是否已超出限制。

表 16. Alert Limit Register (07h) (Read/Write) Description

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	AUL15	AUL14	AUL13	AUL12	AUL11	AUL10	AUL9	AUL8	AUL7	AUL6	AUL5	AUL4	AUL3	AUL2	AUL1	AUL0
POR VALUE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

7.6.9 Manufacturer ID Register (FEh) (Read-Only)

制造商 ID 寄存器 (Manufacturer ID Register) 存储着制造商的唯一识别码

表 17. Manufacturer ID Register (FEh) (Read-Only) Description

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	ID15	ID14	ID13	ID12	ID11	ID10	ID9	ID8	ID7	ID6	ID5	ID4	ID3	ID2	ID1	ID0
POR VALUE	0	1	0	1	0	1	0	0	0	1	0	0	1	0	0	1

ID: **Manufacturer ID Bits** 制造商 ID 位

Bits 0-15

Stores the manufacturer identification bits 存储制造商识别位

7.6.10 Die ID Register (FFh) (Read-Only)

芯片 ID 寄存器存储芯片的唯一识别号和修订 ID。

表 18. Die ID Register (FFh) (Read-Only) Description

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	DID11	DID10	DID9	DID8	DID7	DID6	DID5	DID4	DID3	DID2	DID1	DID0	RID3	RID2	RID1	RID0
POR VALUE	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

DID:
Device ID Bits

Bits 4-15

Stores the device identification bits 存储器件识别位

RID:
Die Revision ID Bits

Bit 0-3

Stores the device revision identification bits 存储器件版本识别位

8 Application and Implementation 应用与实现

注

以下应用部分的信息不属于 TI 组件的技术规格，德州仪器 (TI) 不对其准确性或完整性做出保证。TI 的客户有责任确定组件是否适合其应用目的。请验证并测试设计实现，以确认系统功能。

8.1 Application Information

INA226 是一款具有 I²C™ 兼容接口的电流分流器和功率监视器。该器件可监测分流电阻上的电压降和总线电源电压。可编程的校准值、转换时间和平均功能，结合内部乘法器，能够直接读出以安培为单位的电流和以瓦特为单位的功率。

8.2 Typical Applications

8.2.1 High-Side Sensing Circuit Application

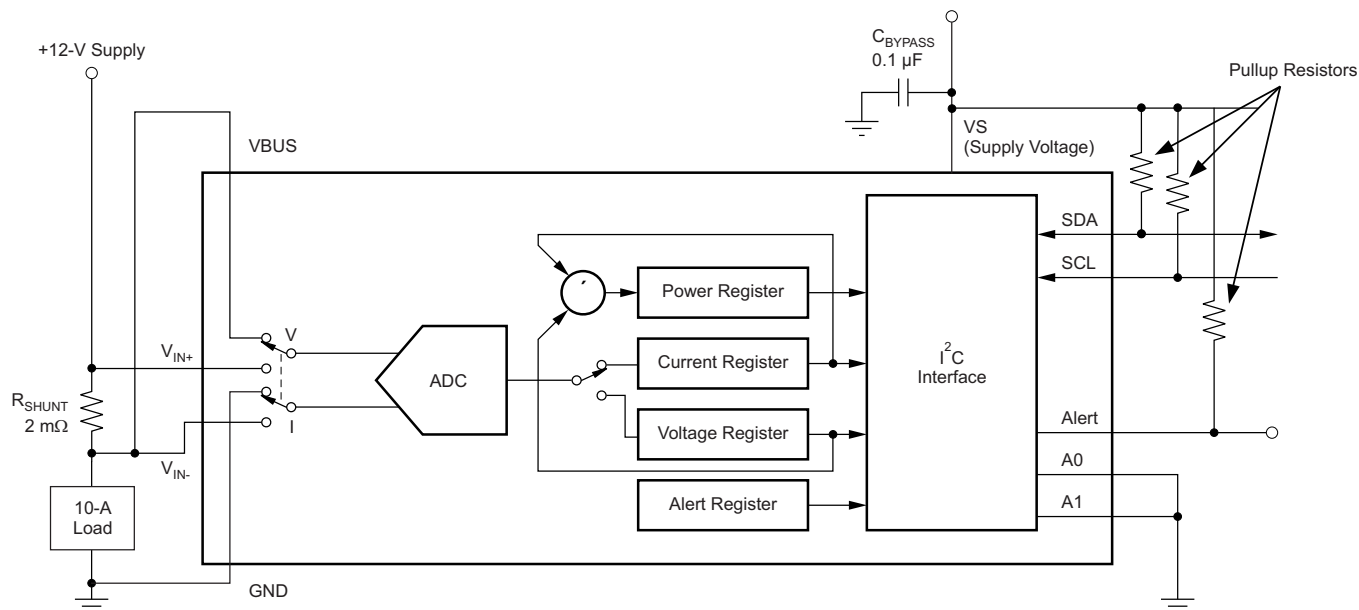


图 27. Typical Circuit Configuration, INA226

8.2.1.1 Design Requirements

INA226 测量电流流过检流电阻 (RSHUNT) 时在其两端产生的电压。该器件还测量总线电源电压，并在校准后能够计算功率。它具有报警功能，报警引脚可以被编程为响应用户定义的事件或转换就绪通知。此设计演示了报警引脚响应设定的阈值的能力。

Typical Applications (接下页)

8.2.1.2 Detailed Design Procedure

The Alert pin can be configured to respond to one of the five alert functions described in the [Alert Pin](#) section. The alert pin must be pulled up to the V_{VS} pin voltage via the pull-up resistors. The configuration register is set based on the required conversion time and averaging. The Mask/Enable Register is set to identify the required alert function and the Alert Limit Register is set to the limit value used for comparison.

8.2.1.3 Application Curves

图 28 shows the Alert pin response to a shunt voltage over-limit of 80 mV for a conversion time (t_{CT}) of 1.1 ms and averaging set to 1. 图 29 shows the response for the same limit but with the conversion time reduced to 140 μ s.

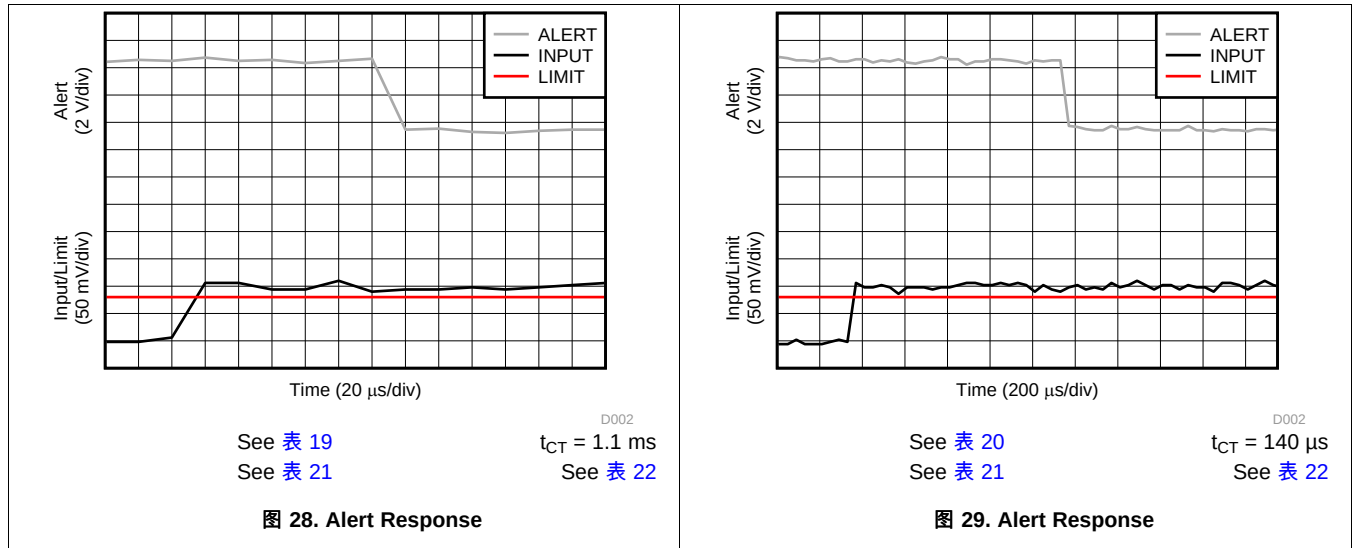


表 19. Configuration Register (00h) Settings for 图 28 (Value = 4025h)

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	RST	—	—	—	AVG2	AVG1	AVG0	VBUSC T2	VBUSC T1	VBUSC T0	V _{SH} CT2	V _{SH} CT1	V _{SH} CT0	MODE3	MODE2	MODE1
POR VALUE	0	1	0	0	0	0	0	0	0	0	1	0	0	1	0	1

表 20. Configuration Register (00h) Settings for 图 29 (Value = 4005h)

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	RST	—	—	—	AVG2	AVG1	AVG0	VBUSC T2	VBUSC T1	VBUSC T0	V _{SH} CT2	V _{SH} CT1	V _{SH} CT0	MODE3	MODE2	MODE1
POR VALUE	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	1

表 21. Mask/Enable Register (06h) Settings for 图 28 and 图 29 (Value = 8000h)

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	SOL	SUL	BOL	BUL	POL	CNVR	—	—	—	—	—	AFF	CVRF	OVF	APOL	LEN
POR VALUE	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

表 22. Alert Limit Register (07h) Settings for 图 28 and 图 29 (Value = 7D00)

BIT #	D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0
BIT NAME	AUL15	AUL14	AUL13	AUL12	AUL11	AUL10	AUL9	AUL8	AUL7	AUL6	AUL5	AUL4	AUL3	AUL2	AUL1	AUL0
POR VALUE	0	1	1	1	1	1	0	1	0	0	0	0	0	0	0	0

9 Power Supply Recommendations

The input circuitry of the device can accurately measure signals on common-mode voltages beyond its power supply voltage, V_{VS} . For example, the voltage applied to the V_{VS} power supply terminal can be 5 V, whereas the load power-supply voltage being monitored (the common-mode voltage) can be as high as 36 V. Note also that the device can withstand the full 0-V to 36-V range at the input terminals, regardless of whether the device has power applied or not.

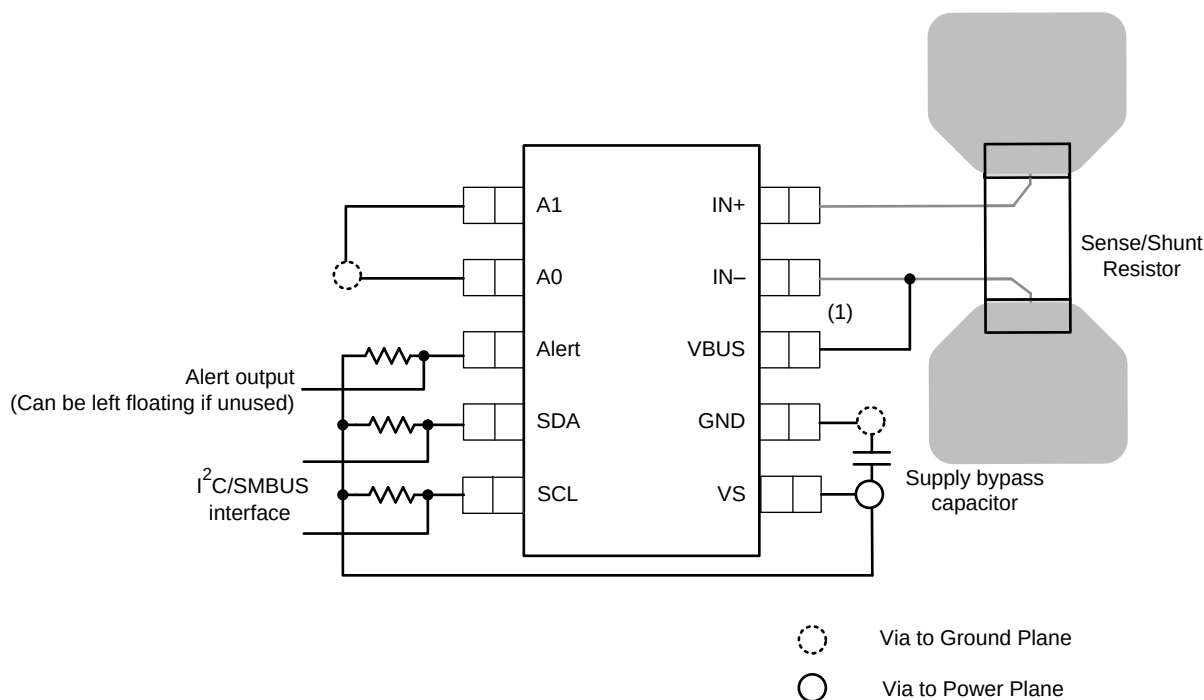
Place the required power-supply bypass capacitors as close as possible to the supply and ground terminals of the device to ensure stability. A typical value for this supply bypass capacitor is 0.1 μF . Applications with noisy or high-impedance power supplies may require additional decoupling capacitors to reject power-supply noise.

10 Layout

10.1 Layout Guidelines

Connect the input pins (IN+ and IN-) to the sensing resistor using a Kelvin connection or a 4-wire connection. These connection techniques ensure that only the current-sensing resistor impedance is detected between the input pins. Poor routing of the current-sensing resistor commonly results in additional resistance present between the input pins. Given the very low ohmic value of the current-sensing resistor, any additional high-current carrying impedance causes significant measurement errors. Place the power-supply bypass capacitor as close as possible to the supply and ground pins.

10.2 Layout Example



(1) connect the VBUS pin to the power supply rail.

图 30. INA226 Layout Example

11 器件和文档支持

11.1 器件支持

11.1.1 开发支持

- 《INA226EVM 评估板和软件教程》（文献编号：SBOU113）

11.2 社区资源

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

TI E2E™ Online Community *TI's Engineer-to-Engineer (E2E) Community*. Created to foster collaboration among engineers. At e2e.ti.com, you can ask questions, share knowledge, explore ideas and help solve problems with fellow engineers.

Design Support *TI's Design Support* Quickly find helpful E2E forums along with design support tools and contact information for technical support.

11.3 商标

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All other trademarks are the property of their respective owners.

11.4 静电放电警告



这些装置包含有限的内置 ESD 保护。存储或装卸时，应将导线一起截短或将装置放置于导电泡棉中，以防止 MOS 门极遭受静电损伤。

11.5 Glossary

[SLYZ022](#) — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

12 机械、封装和可订购信息

以下页中包括机械、封装和可订购信息。 这些信息是针对指定器件可提供的最新数据。 这些数据会在无通知且不
对本文档进行修订的情况下发生改变。 欲获得该数据表的浏览器版本，请查阅左侧的导航栏。

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
INA226AIDGSR	ACTIVE	VSSOP	DGS	10	2500	RoHS & Green	NIPDAUAG SN	Level-2-260C-1 YEAR	-40 to 125	226	Samples
INA226AIDGST	ACTIVE	VSSOP	DGS	10	250	RoHS & Green	NIPDAUAG SN	Level-2-260C-1 YEAR	-40 to 125	226	Samples

⁽¹⁾ The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBsolete: TI has discontinued the production of the device.

⁽²⁾ **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".

RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.

Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

⁽³⁾ MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

⁽⁴⁾ There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

⁽⁵⁾ Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

⁽⁶⁾ Lead finish/Ball material - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

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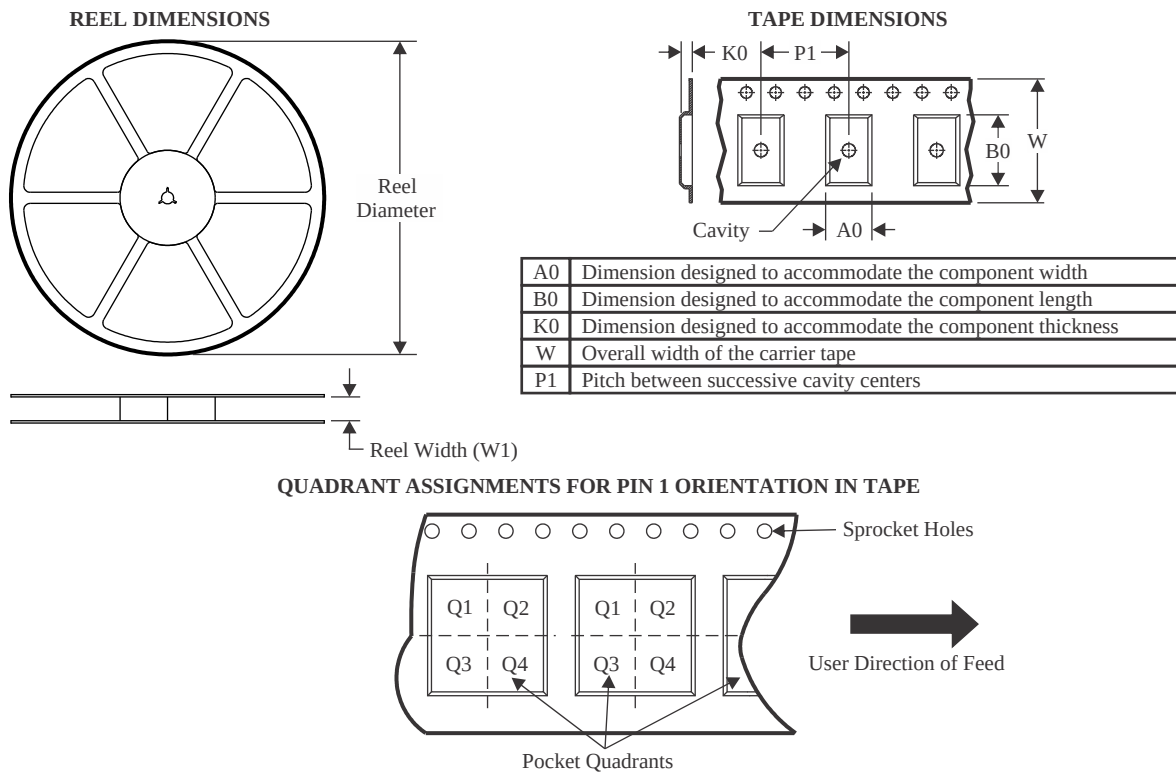
OTHER QUALIFIED VERSIONS OF INA226 :

- Automotive : [INA226-Q1](#)

NOTE: Qualified Version Definitions:

- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects

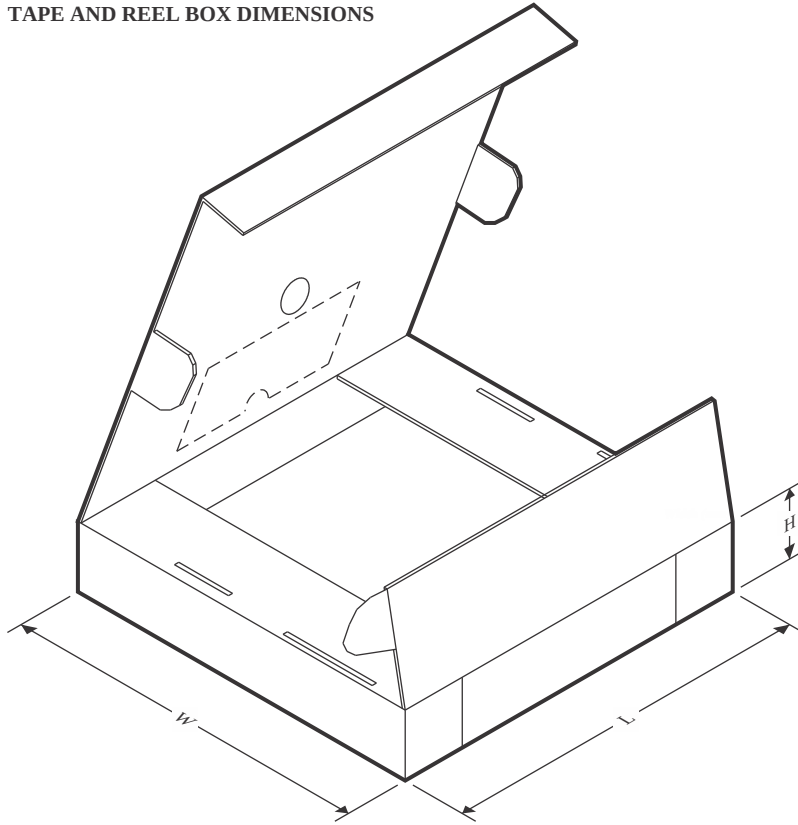
TAPE AND REEL INFORMATION



*All dimensions are nominal

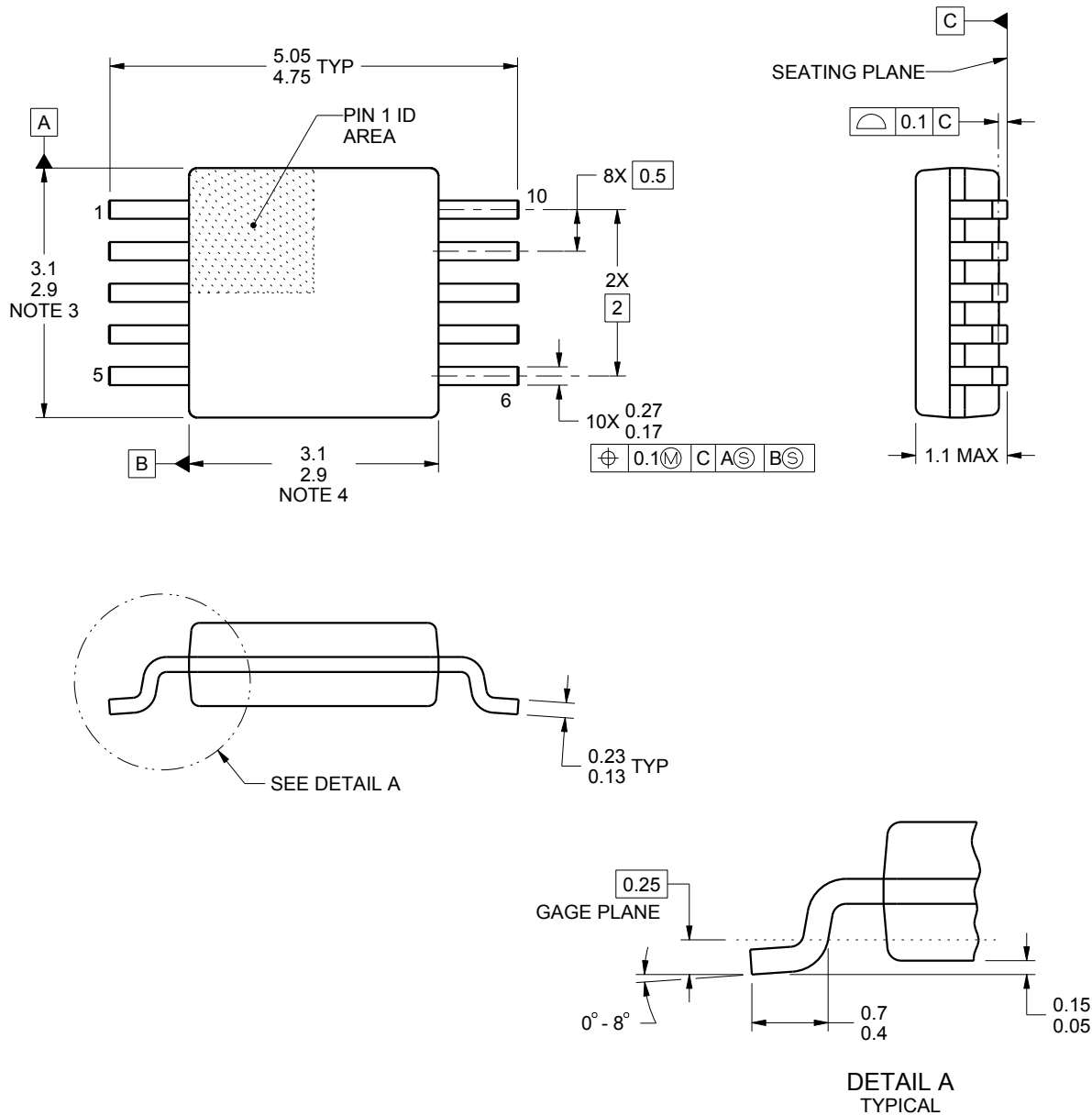
Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
INA226AIDGSR	VSSOP	DGS	10	2500	330.0	12.4	5.3	3.4	1.4	8.0	12.0	Q1
INA226AIDGST	VSSOP	DGS	10	250	330.0	12.4	5.3	3.4	1.4	8.0	12.0	Q1

TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
INA226AIDGSR	VSSOP	DGS	10	2500	366.0	364.0	50.0
INA226AIDGST	VSSOP	DGS	10	250	366.0	364.0	50.0



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NOTES:

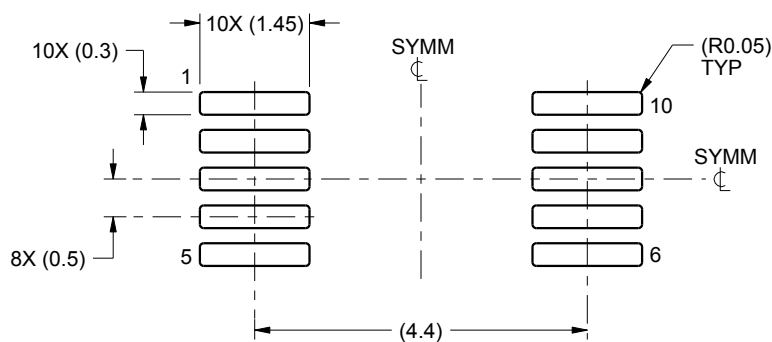
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-187, variation BA.

EXAMPLE BOARD LAYOUT

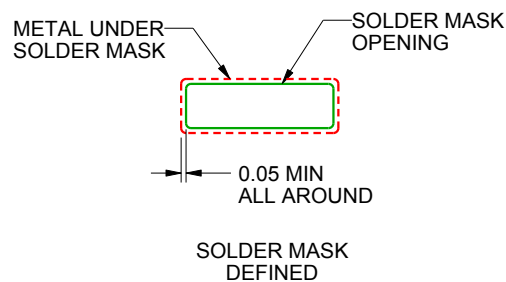
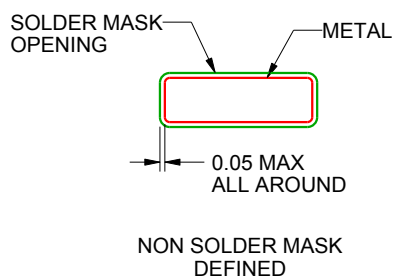
DGS0010A

VSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE
SCALE:10X



SOLDER MASK DETAILS
NOT TO SCALE

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NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

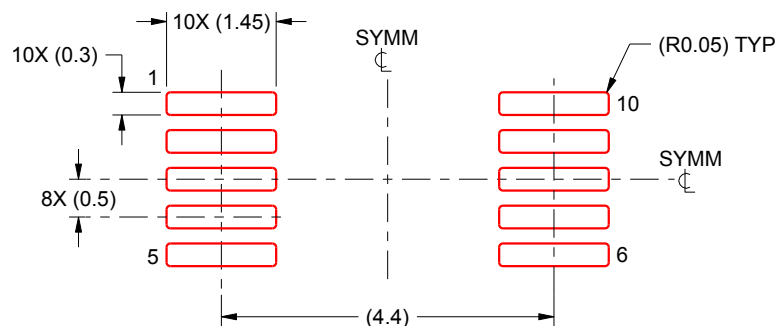
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

DGS0010A

VSSOP - 1.1 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:10X

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NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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